# Fuzzy Logic Based Washing Machine System

## 1. Introduction

Fuzzy logic is a method used to handle uncertainty and approximate reasoning. Unlike traditional logic, it allows values between 0 and 1, making it suitable for real-world problems.

In this project, a fuzzy logic system is designed to determine the washing machine cycle time based on input conditions.

## 2. Objective

The objective of this experiment is:

- To design a fuzzy logic system for washing machine cycle time
- To define fuzzy sets for inputs and output
- To create fuzzy rules
- To implement the system using MATLAB Fuzzy Logic Toolbox

## 3. Inputs and Output

**Inputs:**

- Dirt Level (0–10)
- Load Size (0–10)

**Output:**

- Washing Time (0–60 minutes)

## 4. Fuzzy Sets

### 4.1 Dirt Level

- Low
- Medium
- High

### 4.2 Load Size

- Small
- Medium

- Large

### 4.3 Washing Time

- Short
- Medium
- Long

## 5. Membership Functions

Triangular and trapezoidal membership functions are used to represent fuzzy sets.

- Dirt and Load range: 0–10
- Time range: 0–60

## 6. Fuzzy Rule Base

The following rules are used:

1. IF Dirt is Low AND Load is Small → Time is Short
2. IF Dirt is Medium AND Load is Small → Time is Medium
3. IF Dirt is High AND Load is Small → Time is Long
4. IF Dirt is Low AND Load is Medium → Time is Short
5. IF Dirt is Medium AND Load is Medium → Time is Medium
6. IF Dirt is High AND Load is Medium → Time is Long
7. IF Dirt is Low AND Load is Large → Time is Medium
8. IF Dirt is Medium AND Load is Large → Time is Long
9. IF Dirt is High AND Load is Large → Time is Long

## 7. MATLAB Implementation

The system is implemented using MATLAB Fuzzy Logic Toolbox.

Steps:

- Created Mamdani FIS
- Defined input/output variables
- Assigned membership functions
- Added rules

- Evaluated output using rule viewer

## 8. Results

**Sample Outputs:**

- Dirt = 2, Load = 2 → Time ≈ 7.62 minutes
- Dirt = 5, Load = 5 → Time ≈ 30 minutes
- Dirt = 8, Load = 9 → Time ≈ 52.38 minutes

The system successfully produces realistic washing times based on input conditions.

## 9. Screenshots: The following figures show the membership functions, rule viewer, and output of the fuzzy logic system.

- Dirt Membership Function

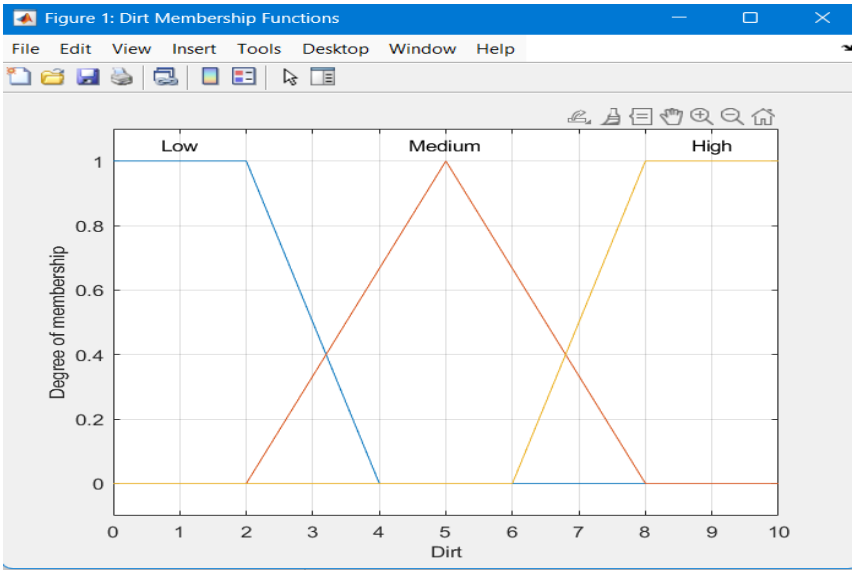


Fig1: Dirt Membership Function

- Load Membership Function

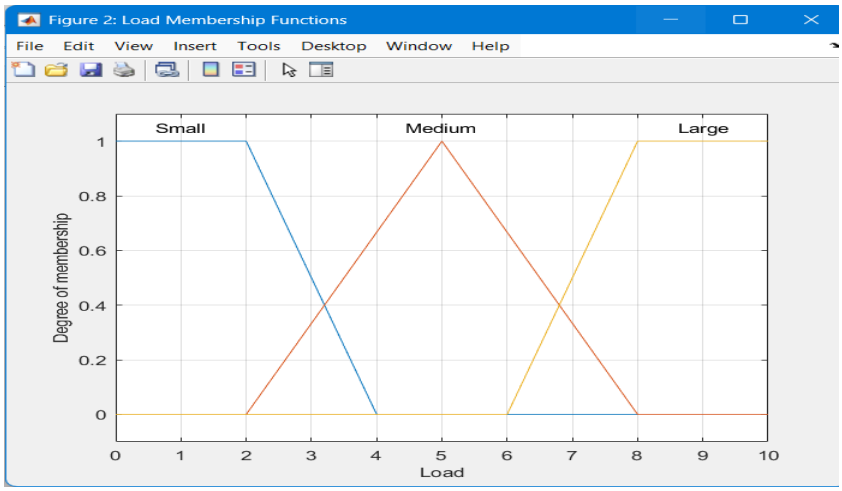


Fig2: Load Membership Function

- Time Membership Function

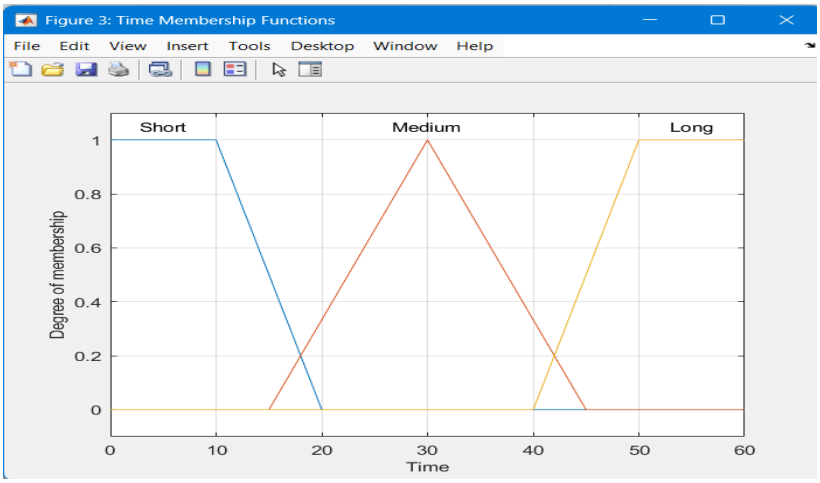


Fig3: Time Membership Function

- Rule Viewer

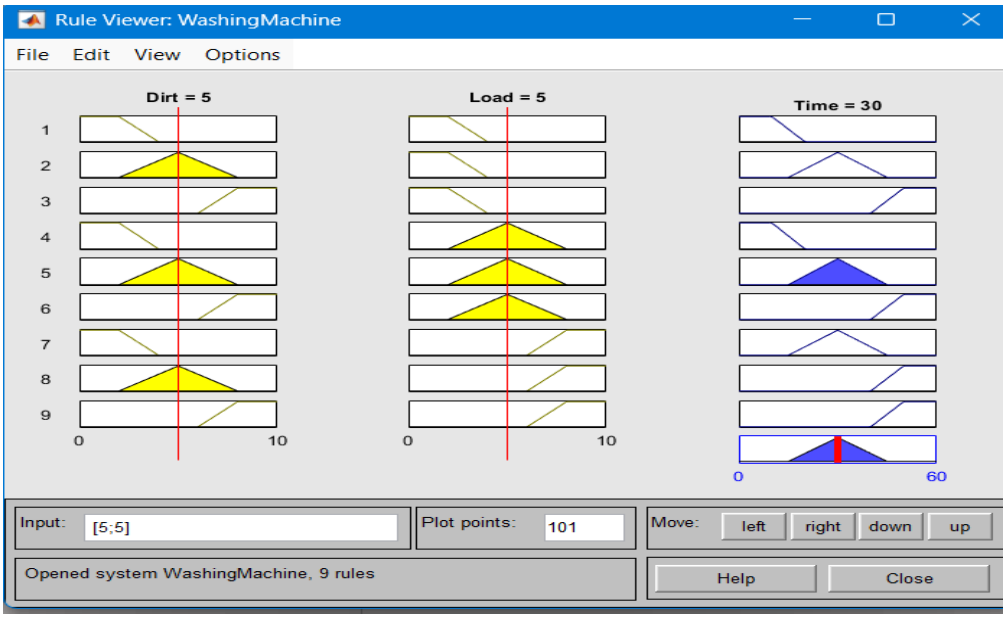


Fig4: Rule Viewer

- Output

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Command Window

ans =

 9×69 char array

  '1. If (Dirt is Low) and (Load is Small) then (Time is Short) (1)     '
  '2. If (Dirt is Medium) and (Load is Small) then (Time is Medium) (1) '
  '3. If (Dirt is High) and (Load is Small) then (Time is Long) (1)     '
  '4. If (Dirt is Low) and (Load is Medium) then (Time is Short) (1)    '
  '5. If (Dirt is Medium) and (Load is Medium) then (Time is Medium) (1)'
  '6. If (Dirt is High) and (Load is Medium) then (Time is Long) (1)    '
  '7. If (Dirt is Low) and (Load is Large) then (Time is Medium) (1)    '
  '8. If (Dirt is Medium) and (Load is Large) then (Time is Long) (1)   '
  '9. If (Dirt is High) and (Load is Large) then (Time is Long) (1)     '

------------- TEST OUTPUTS -------------
Input (Dirt=2.0, Load=2.0) --> Time = 7.62 minutes
Input (Dirt=5.0, Load=5.0) --> Time = 30.00 minutes
Input (Dirt=8.0, Load=9.0) --> Time = 52.38 minutes
```

Fig5: Output

## 10. Conclusion

The fuzzy logic system successfully models washing machine behaviour. It uses linguistic variables and rule-based reasoning to determine washing time efficiently. This approach is flexible and can be extended to other real-world systems.